

mlxconfig Commands

mlxconfig Backup Command

The backup command is used to save the current non-volatile configurations (TLV) in the device into a file in raw TLV syntax so it can be restored anytime using the set_raw command.

mlxconfig backup command allows backing up the all of the configurations which are related only to the PCI Physical Function associated with the given MST device. To back up all of the device configurations, perform the operation from every PCI Physical Function the device exposes. Restoring the configurations must be made from the matching PCI Physical Function.

In a MultiHost environment, these operations are required to be executed per host.

```
1  # mlxconfig -d /dev/mst/mt4117_pciconf0 -f /tmp/backup.conf backup
2  Collecting...
3  Saving output...
4  Done!
5  # cat /tmp/backup.conf
6  MLNX_RAW_TLV_FILE
7  % TLV Type: 0x00000400, Writer ID: ICMD MLXCONFIG(0x09), Writer Host ID: 0x0
8  # mlxconfig -d /dev/mst/mt4117_pciconf0 -f /tmp/backup.conf set_raw
9  Raw TLV #1 Info:
10 Length: 0x14
11 Version: 0
12 OverrideEn: 0
13 Type: 0x00000400
14 Data: 0x00000000 0x000e0000 0x001000f6 0x20160526 0x11250000
15 Operation intended for advanced users.
16 Are you sure you want to apply raw TLV file? ? (y/n) [n] : y
17 Applying... Done!
18 -I- Please reboot machine to load new configurations.
```

Generating an XML Template for the Configurations

Users can generate an XML file that contains a template for the configurations. The template describes the configurations and their parameters. No values are included in the template.

To generate such a template, run the gen_tlvs_file command. This command will generate a file containing a list of all supported configurations by mlxconfig, with a zero appearing in the

end of each configuration. To choose a configuration, change the 0 to 1, then save the file and run the `gen_xml_template` command. An XML file containing the required configurations will be generated.

Example:

```
1  # mlxconfig gen_tlvs_file /tmp/confs.txt
2  Saving output...
3  Done!
4  # cat /tmp/confs.txt
5  nv_host_to_bmc      0
6  nv_kdnet_data       0
7  nv_fpga_data        0
8  nv_packet_pacing    0
9  nv_debug_mode       0
10 nv_global_pci_conf  0
```

In order to include the `nv_kdnet_data` configuration in the template, change the 0 to 1, as demonstrated in the following example.

Example:

```
1  nv_kdnet_data      1
2
3  #mlxconfig gen_xml_template /tmp/confs.txt /tmp/template.xml
4  Saving output...
5  Done!
6
7  #cat /tmp/template.xml
8  <?xml version="1.0" encoding="UTF-8"?>
9  <config xmlns="http://www.mellanox.com/config">
10 <nv_kdnet_data>
11
12         <!-- Legal Values: False/True -->
13         <kdnet_en></kdnet_en>
14
15 </nv_kdnet_data>
16 </config>
```

mlxconfig xml2raw Command

The `xml2raw` command is an easy way to generate a flawless raw configuration file that can be used in the `set_raw` command. The input for the command is an XML file that contains the data of the required configurations. To generate an XML file and fill it with the desired values, run the commands from [Generating an XML Template for the Configurations](#), and then use the `xml2raw` command to generate a raw file.

Example:

```
1 # cat /tmp/template.xml
2 <?xml version="1.0" encoding="UTF-8"?>
3 <config xmlns="http://www.mellanox.com/config">
4 <nv_kdnet_data>
```



```
8
9 </nv_kdnet_data>
10 </config>
11
12 #mlxconfig xml2raw /tmp/template.xml /tmp/confs.raw
13 Saving output...
14 Done!
15
16 #cat /tmp/confs.raw
17 MLNX_RAW_TLV_FILE
18 0x03000004 0x00000085 0x00000000 0x80000000
```

mlxconfig xml2bin Command

The xml2bin command is an easy way to generate a binary file that contains a binary dump of configurations. The input for the command is an XML file that contains the data of the required configurations. To generate an XML file and fill it with the desired values, run the commands from Section 2.4.10, and then use the xml2bin command to generate a binary file.

Example:

```
1 #mlxconfig xml2bin /tmp/template.xml /tmp/confs.bin
2 Saving output...
3 Done!
4 # hexdump -C /tmp/confs.bin
5 00000000 80 00 00 00 |....|
6 00000004
```

mlxconfig create_conf Command

The create_conf command assists in creating an NV configuration file that can be used for LifeCycle and Secure Firmware Updates purposes. The flow for creating a configuration file is the same as the flow for xml2bin. The user must provide the command with an XML file containing the required configurations and their values. The command can sign the configuration file, if the user provides a private key and UUID. The sign result will be appended to the end of the configuration file. If no private key and UUID are provided, the tool will compute an SHA256/SHA512 digest and append it to the file. The generated signature will be used by the firmware for authentication purposes.

Currently the only supported NV configurations types are CS tokens, Debug tokens, BTC tokens, and MLNX ID which are used for Secure Firmware Updates. Additionally, it supports two RSA keys of 2048 or 4096 bits length for all tokens except BTC which supports only 4096 bits.

The NV configurations files must have the applicable_to configuration.

Examples:

```
1 # mlxconfig create_conf --private_key privatekey.pem --key_uuid "29ee36ee-13
2 Saving output...
3 Done!
```

```
1 # mlxconfig create_conf --private_key privatekey.pem --key_uuid "29ee36ee-13
2 Saving output...
3 Done!
```

mlxconfig apply Command

The apply command can be used to apply the NV configurations files to the firmware using the apply command. Only Firmware that supports applying configurations files can be used.

Example:

```
1 # mlxconfig -d /dev/mst/mt4115_pciconf0 apply /tmp/nvconf.bin
2 Applying...
3 Done!
```

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